

Question 14

1 mark

Identify the maximum number of x -intercepts for a polynomial function of degree 3.

- a) 1
- b) 2
- c) 3
- d) 4

Question 15

1 mark

The graph of $y = f(x)$ contains the point (a, b) . The graph of $g(x)$ is a transformation of the graph of $f(x)$ and contains the point $(3a, b)$.

Identify the function that represents $g(x)$.

- a) $g(x) = f(3x)$
- b) $g(x) = 3f(x)$
- c) $g(x) = f\left(\frac{x}{3}\right)$
- d) $g(x) = \frac{1}{3}f(x)$

Question 16

1 mark

The angle 2.95 radians, in standard position, terminates in quadrant:

- a) I
- b) II
- c) III
- d) IV

Question 17

1 mark

Evaluate:

$$2 \sin \frac{\pi}{8} \cos \frac{\pi}{8}$$

a) $\frac{1}{2}$

b) $\frac{\sqrt{2}}{2}$

c) 1

d) $\sqrt{2}$

Question 18

1 mark

Identify which of the following represents the 5th term in the expansion of $(4x^2 - 2y^3)^{15}$.

a) ${}_{15}C_5 (4x^2)^{10} (-2y^3)^5$

b) ${}_{15}C_5 (4x^2)^{11} (-2y^3)^4$

c) ${}_{15}C_4 (4x^2)^{10} (-2y^3)^5$

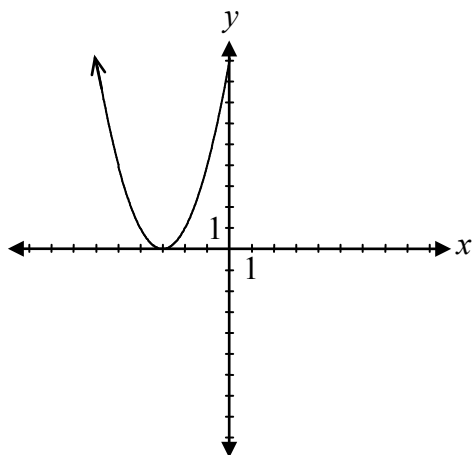
d) ${}_{15}C_4 (4x^2)^{11} (-2y^3)^4$

Question 19

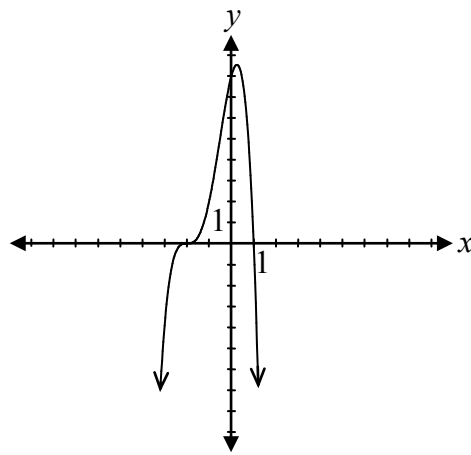
1 mark

Identify which of the following graphs of polynomial functions has a zero with a multiplicity of 3.

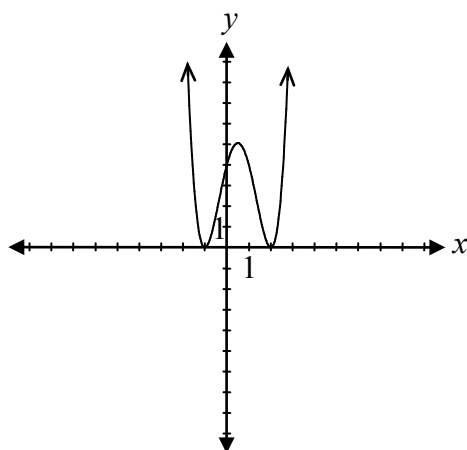
a)



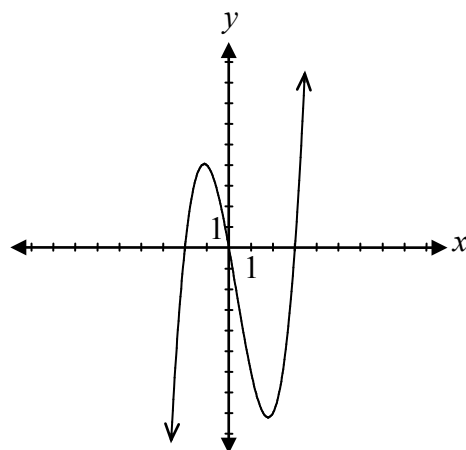
b)



c)



d)



Question 20

1 mark

A non-permissible value of x for the function $f(x) = \frac{1}{\cos x + 1}$ is:

- a) -1
- b) 0
- c) π
- d) $\frac{3\pi}{2}$

Question 21

1 mark

Identify which of the following statements is true for the rational function $f(x) = \frac{4(x-1)(x-2)}{(x-1)(x+3)}$.

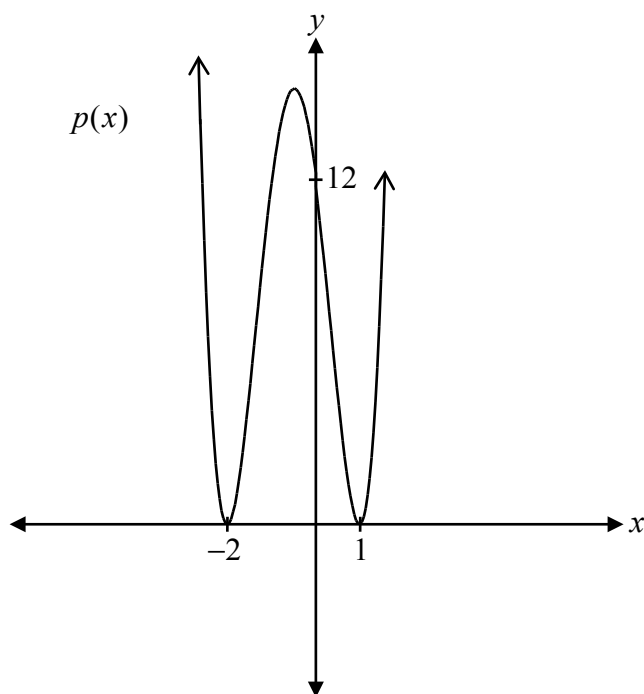
- a) The equation of the horizontal asymptote is $y = 4$.
- b) The equation of the vertical asymptote is $x = 1$.
- c) The y -intercept is 0 .
- d) There is a point of discontinuity (hole) when $x = 2$.

Question 22

3 marks

115

Determine the equation of the polynomial function, $p(x)$, represented by the graph.



$p(x) =$ _____

Question 23

1 mark

116

Evaluate:

$$\log_4 2$$

Question 24

2 marks

117

Evaluate:

$$\left(\cos \frac{11\pi}{3}\right)\left(\csc \frac{11\pi}{6}\right)$$

Question 25

1 mark

118

Estimate the value of $\log_2 5$.

Justify your answer.

Question 26

2 marks

119

If θ terminates in quadrant III and $\cos \theta = -\frac{6}{7}$, determine the exact value of $\tan \theta$.

Question 27

1 mark

120

Given $f(x) = x^2 + x - 4$ and $g(x) = \sqrt{x+5}$, Taz was asked to find $f(g(x))$.

Taz's solution:

$$\begin{aligned} f(g(x)) &= (\sqrt{x+5})^2 + x - 4 \\ &= x + 5 + x - 4 \\ &= 2x + 1, \quad x \geq -5 \end{aligned}$$

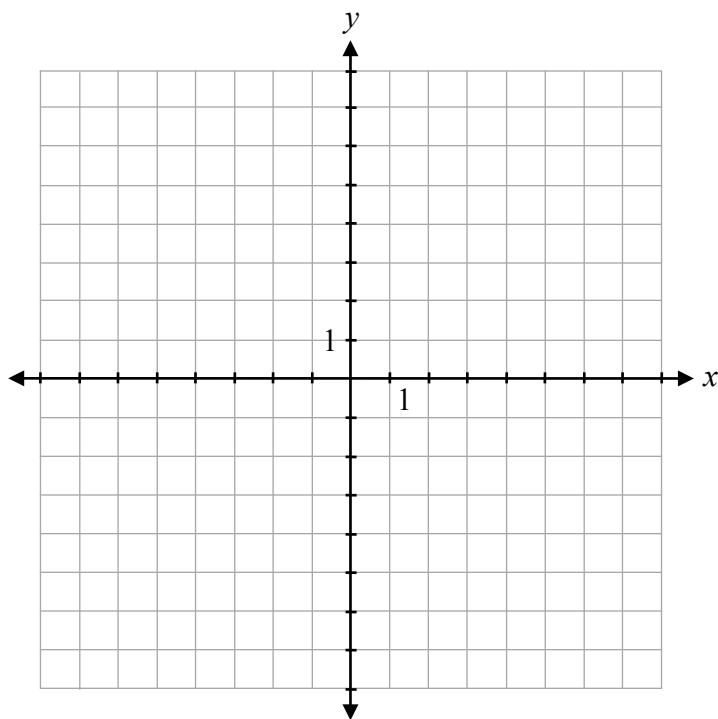
Describe the error in Taz's solution.

Question 28

3 marks

121

Sketch the graph of the function $f(x) = 3 \log_2(x+1)$.



Question 29

1 mark

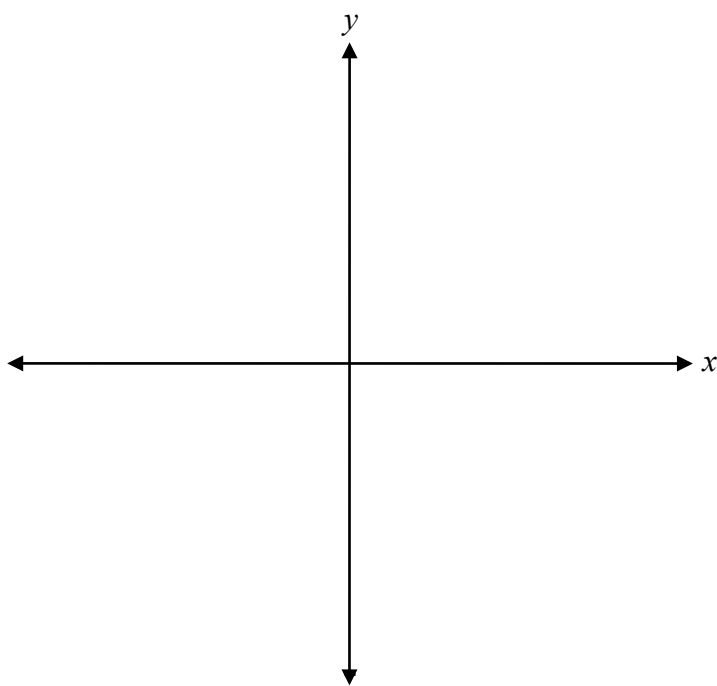
122

Write an equation of a rational function that would not have any vertical asymptotes.

Determine the exact value of $\tan 75^\circ$.

Sketch the graph of the following function:

$$f(x) = \frac{(x+3)(x-3)}{x(x-3)}$$



In the binomial expansion of $\left(\frac{1}{x^3} - 2x^2\right)^9$, determine which term contains x^3 .

Question 33

4 marks

126

José and Dana get on a Ferris wheel, which is 1 metre off the ground. The diameter of the Ferris wheel is 16 metres. Their ride lasts for 4 minutes, in which time the Ferris wheel makes one revolution.

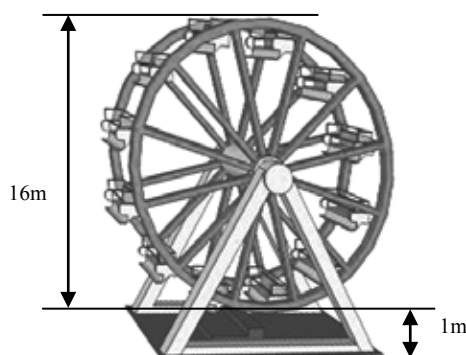
Determine the values of A , B , C , and D , if the sinusoidal function that models the situation is $h(t) = A \cos[B(t - C)] + D$, where h is the height at which José and Dana are located on the Ferris wheel, from the ground, in metres, and t is the time, in minutes.

$A =$ _____

$B =$ _____

$C =$ _____

$D =$ _____



Question 34

3 marks

127

Solve algebraically:

$${}_nP_3 = 4!(n-1)$$

Question 35

2 marks

128

Given $f(x) = \frac{2}{x-1}$, determine the equation of the inverse, $f^{-1}(x)$.

Question 36

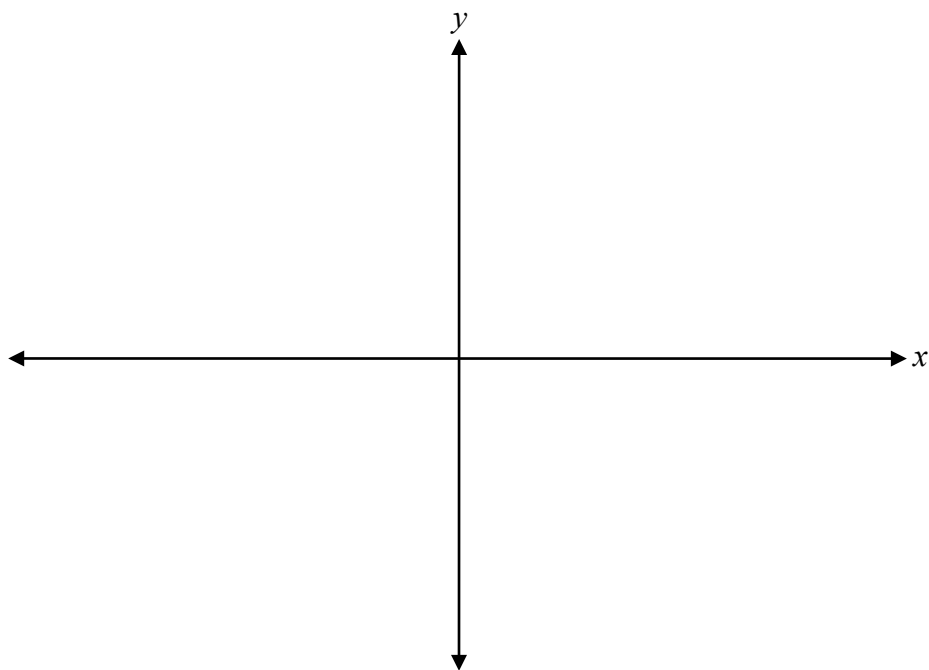
3 marks

129

Solve:

$$4\log_3 2 - \frac{1}{3}\log_3 8 = \log_3 a$$

Sketch the graph of at least one period of the function $y = 3 \cos(\pi x) - 1$.



Question 38

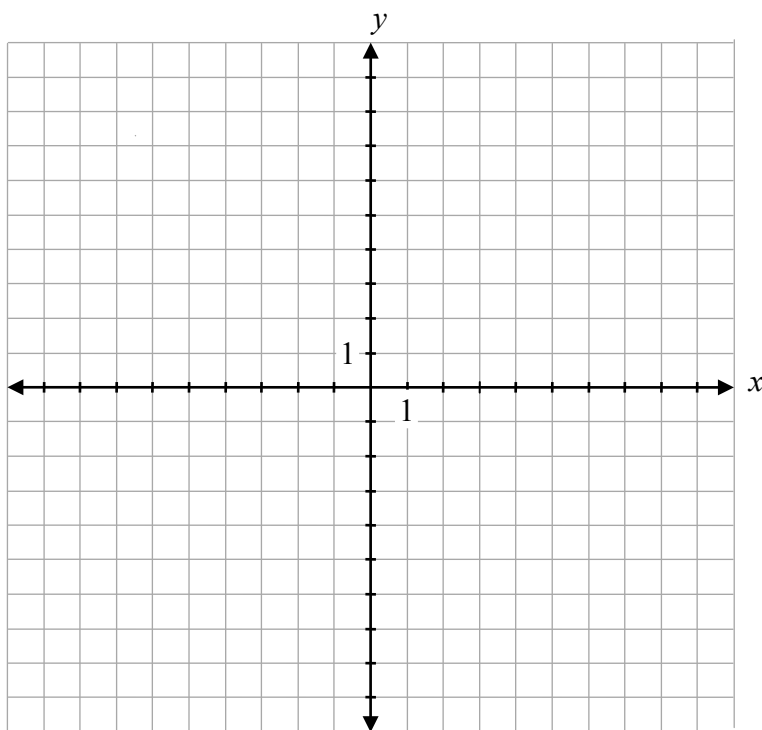
3 marks

131

Using the laws of logarithms, fully expand the expression:

$$\log_a \left(\frac{x^3}{y\sqrt{z}} \right)$$

Sketch the graph of $f(x) = 3\sqrt{x-2} + 1$.



Question 40

a) 1 mark b) 1 mark

133
134

a) Determine the domain of the graph of the function $f(x) = \sqrt{x^2 - 4}$.

b) Explain why the domain of $f(x) = \sqrt{x^2 - 4}$ is restricted.

Question 41

a) 1 mark b) 1 mark

135
136

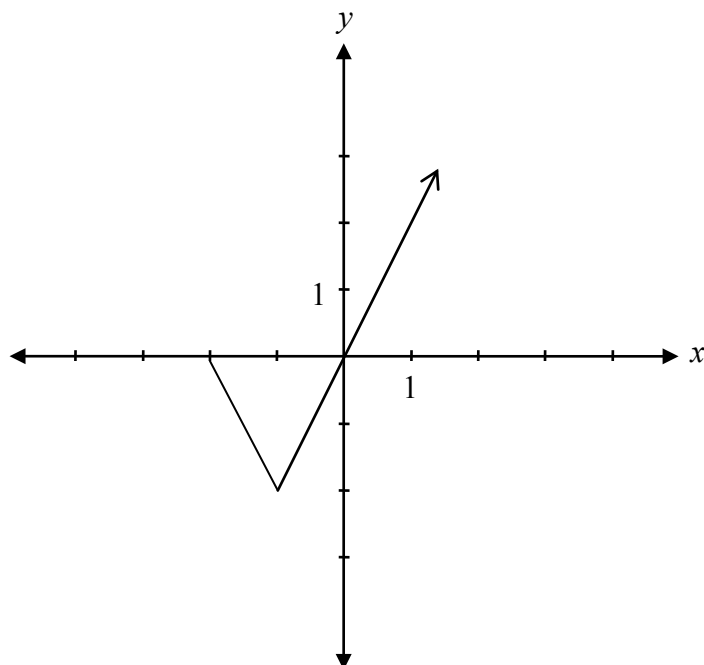
Given the point $(-12, -18)$ on the graph of $f(x)$, determine the new points after the following transformations of $f(x)$.

a) $\frac{1}{f(x)}$

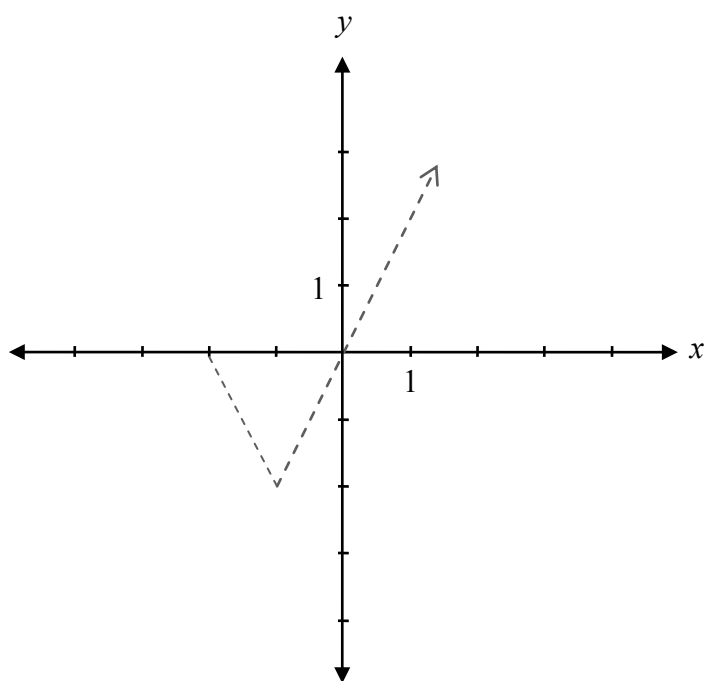
b) $f(-x) + 10$

Explain why there is no solution for the equation $\csc \theta = -\frac{1}{2}$.

Given the graph of $y = f(x)$,



sketch the graph of $y = |f(2x)| + 1$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 44

1 mark

139

Given $f(x) = 2^x + 1$, state the equation of the horizontal asymptote.

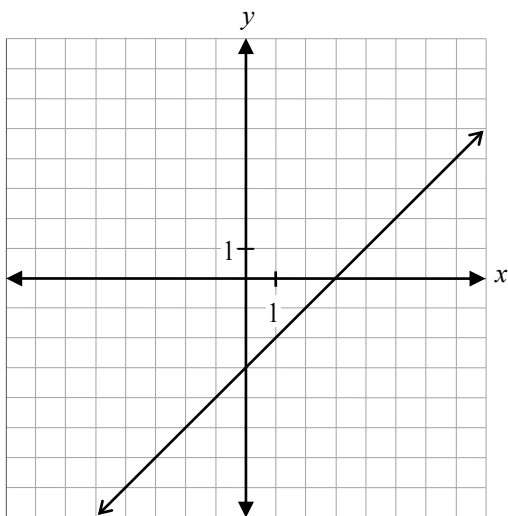
Question 45

2 marks

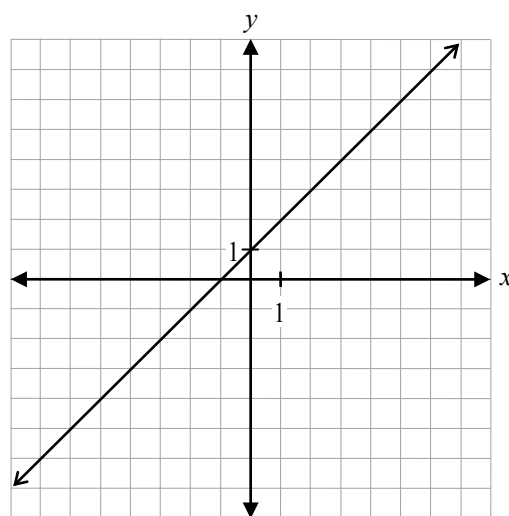
140

Given the following graphs of $f(x) = x - 3$ and $g(x) = x + 1$,

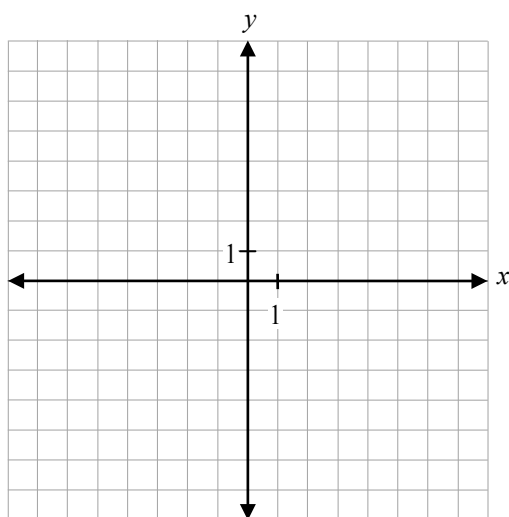
$f(x)$



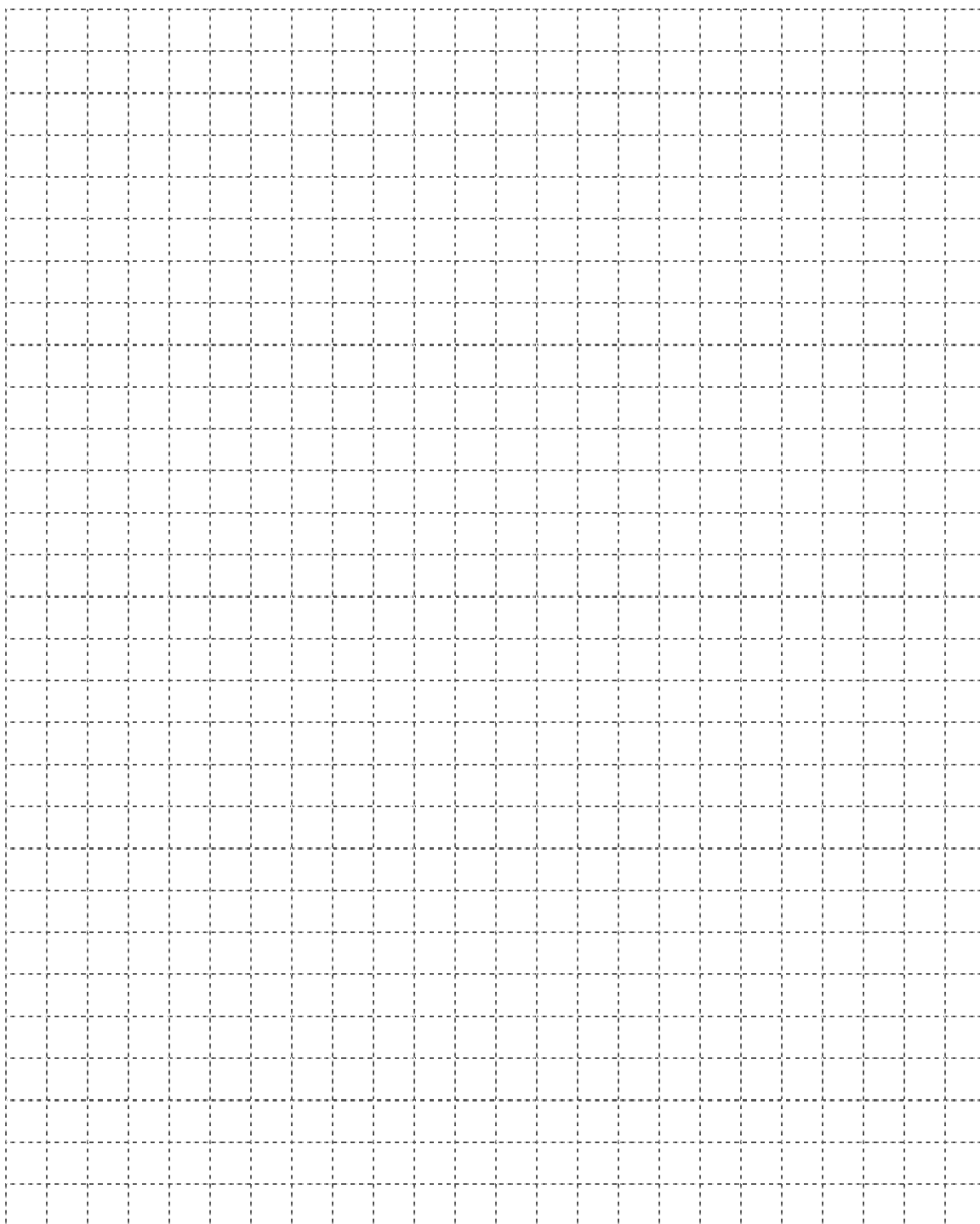
$g(x)$



sketch the graph of $h(x) = (f \cdot g)(x)$.



No marks will be awarded for work done on this page.



No marks will be awarded for work done on this page.